

ZHOU ZHANG

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Education

- **Beihang University** 09/2023 – 06/2027 (expected), Beijing, China
Ph.D. in Cyber Science and Technology Supervisors: Zhenyu Guan and Song Bian
- **Beihang University** 09/2021 - 06/2023, Beijing, China
M.S. in Cyber Science and Technology Supervisor: Zhenyu Guan
- **Beihang University** 09/2017 - 06/2021, Beijing, China
B.S. in Cyber Science and Technology

Research Interests

- Fully Homomorphic Encryption
- Private Information Retrieval
- Encrypted Databases
- Privacy-Preserving Computation

Publications

- **Zhou Zhang**, Ran Mao, Zian Zhao, Haowen Pan, Yunhao Fu, Yizhong Liu, Dawei Li, Yi Chen, Bo Zhang, Changrui Ren, Jin Dong, Zhenyu Guan, and Song Bian, SPIRIT: Batch Hintless Single-Server PIR via Stateful Ciphertext Conversion, USENIX Security, 2026.
- **Zhou Zhang**, Song Bian, Zian Zhao, Ran Mao, Haoyi Zhou, Jiafeng Hua, Yier Jin, and Zhenyu Guan, ArcEDB: An Arbitrary-Precision Encrypted Database via (Amortized) Modular Homomorphic Encryption, ACM CCS, 2024.
- Song Bian*, **Zhou Zhang***, Haowen Pan, Ran Mao, Zian Zhao, Yier Jin, and Zhenyu Guan, HE3DB: An Efficient and Elastic Encrypted Database Via Arithmetic-And-Logic Fully Homomorphic Encryption, ACM CCS, 2023.
Distinguished Paper Award. (* Equal contribution.)
- Song Bian, Haowen Pan, Jiaqi Hu, **Zhou Zhang**, Yunhao Fu, Jiafeng Hua, Yi Chen, Bo Zhang, Yier Jin, Jin Dong, and Zhenyu Guan, Engorgio: An Arbitrary-Precision Unbounded-Size Hybrid Encrypted Database via Quantized Fully Homomorphic Encryption, USENIX Security, 2025.
- Song Bian, Zian Zhao, **Zhou Zhang**, Ran Mao, Kohei Suenaga, Yier Jin, Zhenyu Guan, and Jianwei Liu, HEIR: A Unified Representation for Cross-Scheme Compilation of Fully Homomorphic Computation, NDSS, 2024.
- Song Bian, Zian Zhao, Ruiyu Shen, **Zhou Zhang**, Ran Mao, Dawei Li, Yizhong Liu, Masaki Waga, Kohei Suenaga, Zhenyu Guan, Jiafeng Hua, Yier Jin, and Jianwei Liu, CHLOE: Loop Transformation over Fully Homomorphic Encryption via Multi-Level Vectorization and Control-Path Reduction, IEEE S&P, 2025.
- Ran Mao, **Zhou Zhang**, Zian Zhao, Zhenyu Guan, Peng Yin, and Song Bian, SALUS: Large-Scale Homomorphic Circuit Synthesis via Logic-Aware LUT Optimization, IEEE Transactions on Information Forensics and Security, 2026.
- Luchang Lei, Yongqing Zhu, Xinhang Zou, **Zhou Zhang**, Zian Zhao, Yifan He, Gangfeng Du, Zhenyu Guan, Huazhong Yang, Yongpan Liu, Song Bian, and Hongyang Jia, Enabling Energy-Efficient Homomorphic Encryption Evaluation via eDRAM-Based In-Situ Computing in an Edge Processor, IEEE Journal of Solid State Circuits, 2025.
- Luchang Lei, Yongqing Zhu, Xinhang Zou, Yifan He, **Zhou Zhang**, Zhenyu Guan, Huazhong Yang, Yongpan Liu, Song Bian, and Hongyang Jia, An eDRAM-Based In-Situ-Computing Processor for Homomorphic Encryption Evaluation on the Edge, ESSERC, 2024.
- Zhenyu Guan, Ran Mao, Qianyun Zhang, **Zhou Zhang**, Zian Zhao, and Song Bian, AutoHoG: Automating Homomorphic Gate Design for Large-Scale Logic Circuit Evaluation, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2024.
- Zhenyu Guan, Junpeng Jing, Xin Deng, Mai Xu, Lai Jiang, **Zhou Zhang**, and Yipeng Li, DeepMIH: Deep Invertible Network for Multiple Image Hiding, IEEE Transactions on Pattern Analysis and Machine Intelligence, 2023.

Preprints

- **Zhou Zhang**, Song Bian, Zhenyu Guan, and Giuseppe Persiano, RotPIR: Sublinear Single-Server PIR with Efficient Pre-Processing via Oblivious Homomorphic Rotation, Cryptology ePrint Archive, Paper 2026/1336, 2026.

Research Funding

- **National Natural Science Foundation of China (NSFC)** 01/2025 – Present
Young Student Basic Research Program; Principal Investigator (PI)
Efficient Bootstrapping for Fully Homomorphic Encryption via Module Lattices
- **Cyberspace Administration of China & Ant Research** 10/2022 – 09/2023
Innovation Funding Program for Schools of Cybersecurity; Principal Investigator (PI)
Privacy-Preserving K-Means and SVM

Honors & Awards

- **Distinguished Paper Award**, ACM CCS 2023 11/2023
- Best Presentation Award, China Cyber Security Congress 2026 09/2026
- Young Talent Support Program for Doctoral Students, China Association for Science and Technology 2024
- National Scholarship (Top 1%), Ministry of Education of China 2024, 2020
- Excellent Student (Top 5%) of Beihang University 2024, 2023, 2018
- Outstanding Graduate (Top 3%) of Beihang University 06/2021

Academic Service

- **Artifact Evaluation Program Committee**
USENIX Security 2026; ACM CCS 2026.
- **Journal Reviewer**
IEEE Transactions on Knowledge and Data Engineering (TKDE)
IEEE Transactions on Dependable and Secure Computing (TDSC)

Presentations

- **HE3DB: An Efficient and Elastic Encrypted Database Via Arithmetic-And-Logic Fully Homomorphic Encryption**
Huawei FHE Seminar, Beijing, China 09/2023
ACM CCS, Copenhagen, Denmark 11/2023
- **ArcEDB: An Arbitrary-Precision Encrypted Database via (Amortized) Modular Homomorphic Encryption**
ACM CCS, Salt Lake City, Utah, USA 10/2024
- **SPIRIT: Batch Hintless Single-Server PIR via Stateful Ciphertext Conversion**
Cybersecurity Conference of the Chinese Institute of Electronics, Changsha, China 06/2026
Crypto Reading Group, Beihang University, Beijing, China 07/2026
- **Querying Encrypted Data with Fully Homomorphic Encryption (Best Presentation Award)**
Doctoral Student Forum, China Cyber Security Congress (CCSC 2026), Hefei, China 09/2026

Teaching Experience

- Teaching Assistant of Cryptographic Applications in Security, Beihang University 03/2024 – 07/2024
- Teaching Assistant of International Engineering Fundamentals, Beihang University 03/2022 – 07/2022
- Teaching Assistant of Linear Algebra, Beihang University 09/2020 – 01/2021

Technical Skills

- **Programming:** C++, Python, SQL, LaTeX
- **Languages:** Chinese, English